



A

Project Report

on

MERN Stack Web Development and Agentic AI Foundations

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By

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ABSTRACT

This report presents an account of the Summer Internship undertaken by Paarth Gupta, a Computer Science and Engineering student of KIET Group of Institutions, Ghaziabad, during the session 2026-27. The internship was pursued through two globally recognized platforms: the Cisco Networking Academy, offering the JavaScript Essentials 1 course under the JS Institute curriculum, and Oracle University, offering the Agentic AI Certified Foundations program.

The internship focused on strengthening core programming logic and modern web application development using JavaScript, and extended into the MERN Stack ecosystem, covering React.js for frontend development and Node.js for backend development. In parallel, the intern gained a practical understanding of Agentic AI concepts, including Large Language Model (LLM) fundamentals, prompt engineering, and modern AI agent frameworks such as LangChain, LangGraph, and Retrieval-Augmented Generation (RAG).

Through this internship, the intern successfully earned two industry-recognized certifications: the Cisco Networking Academy "JavaScript Essentials 1" certification and the Oracle Certified Foundations Associate certification in Agentic AI. The report discusses the objectives, methodology, technology stack, key learnings, challenges encountered, and the overall outcomes of the internship, concluding with the scope for future learning in Full Stack Development and Artificial Intelligence.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form	
AI	Artificial Intelligence	
JS	JavaScript	
MERN	MongoDB, Express.js, React.js, Node.js	
LLM	Large Language Model	
RAG	Retrieval-Augmented Generation	
NAAC	National Assessment and Accreditation Council	
KIET	KIET Group of Institutions	

CHAPTER 1 INTRODUCTION

1.1 Introduction

This report documents the Summer Internship carried out by the author as part of the Bachelor of Technology programme in Computer Science and Engineering at KIET Group of Institutions, Ghaziabad. The internship was completed through two industry-recognized online learning platforms, namely the Cisco Networking Academy and Oracle University, with the objective of gaining hands-on exposure to modern web development technologies and emerging Artificial Intelligence practices.

The Cisco Networking Academy is a global IT and cybersecurity education program run by Cisco Systems that bridges the technology skills gap through an industry-standard curriculum; the JavaScript Essentials 1 course under this program was undertaken to build a strong foundation in core programming logic and application development. Oracle University, a global leader in enterprise cloud computing, databases, and infrastructure, offered the Agentic AI Certified Foundations program, which introduced next-generation AI capabilities and automation concepts through its Agentic AI curriculum.

Table 1.1 below summarizes the two certifications completed as part of this internship.

Certification	Issuing Body	Completion Date
JavaScript Essentials 1	Cisco Networking Academy (JS Institute)	24 Jul 2026
Agentic AI Certified Foundations Associate	Oracle University	04 Aug 2026

1.2 Project Description

The internship was structured around two parallel learning tracks: a Full Stack Web Development track based on JavaScript and the MERN Stack, and an Artificial Intelligence track based on Agentic AI concepts. The specific objectives set out for the internship were as follows:

- Earn industry-recognized professional certifications in JavaScript and Agentic AI.
- Build a strong foundation in JavaScript for modern web development.
- Strengthen skills in the MERN Stack ecosystem.
- Advance frontend development capabilities using React.js.

- Enhance backend development knowledge with Node.js.
- Gain a practical understanding of Agentic AI concepts and workflows.
- Explore emerging AI technologies shaping the future of software development.
- Prepare for industry-ready roles in Full Stack Development and Artificial Intelligence.

CHAPTER 2 LITERATURE REVIEW

2.1 Cisco Networking Academy

The Cisco Networking Academy is a global IT and cybersecurity education initiative run by Cisco Systems. It bridges the technology skills gap by delivering an industry-standard curriculum to learners worldwide. As part of this internship, the JavaScript Essentials 1 course, developed under the JS Institute (Open Education and Development Group), was undertaken. This course focuses on core programming logic and application development using JavaScript, forming the basis for further study in modern web technologies.

2.2 Oracle University

Oracle University, the training and certification arm of Oracle Corporation, is a global leader in enterprise cloud computing, databases, and infrastructure, and has been a pioneer in integrating Artificial Intelligence into enterprise applications.

2.2.1 Agentic AI Certified Foundations Program

The Agentic AI Certified Foundations program offered by Oracle University focuses on next-generation AI capabilities and automation. It introduces learners to the fundamentals of Agentic AI and autonomous AI systems, Large Language Models (LLMs), prompt engineering, and AI agent workflows involving reasoning, planning, and tool integration. The program also provides exposure to modern AI frameworks such as LangChain, LangGraph, and Retrieval-Augmented Generation (RAG), which are widely used in building intelligent, autonomous applications.

CHAPTER 3 PROPOSED METHODOLOGY

3.1 Technology Stack and Learning Approach

The internship followed a structured, progressive learning methodology divided into two complementary tracks, as illustrated in Figure 3.1. In the web development track, learning began with core JavaScript concepts and progressed to React.js for frontend development and Node.js for backend development, culminating in the MERN Stack for industry-level web development.

In the Artificial Intelligence track, the methodology began with Agentic AI fundamentals offered by Oracle University, moving through LLM fundamentals and prompt engineering, and advancing to modern agentic frameworks such as LangChain, LangGraph, and Retrieval-Augmented Generation (RAG), before culminating in AI agent development.

Web Development Track (JavaScript)	Artificial Intelligence Track (Agentic AI)
JavaScript	Agentic AI (Oracle University)
React.js and Node.js	LLM Fundamentals
MERN Stack	Prompt Engineering
Industry Level Web Development	LangChain, LangGraph, RAG
	AI Agent Development

Figure 3.1: Technology Stack adopted during the Internship

This dual-track methodology allowed practical, hands-on coding exercises to be combined with conceptual study of AI agent workflows, enabling the intern to connect theoretical concepts with real-world application development.

CHAPTER 4 RESULTS AND DISCUSSION

4.1 Key Learnings

Technical Skills

- Developed a strong foundation in JavaScript and modern programming concepts.
- Strengthened understanding of React.js for component-based frontend development.
- Enhanced backend development knowledge using Node.js and asynchronous programming concepts.
- Improved problem-solving through practical coding exercises.

Artificial Intelligence

- Learned the fundamentals of Agentic AI and autonomous AI systems.
- Explored Large Language Models (LLMs) and prompt engineering.
- Understood AI agent workflows, including reasoning, planning, and tool integration.
- Gained exposure to modern AI frameworks such as LangChain, LangGraph, and Retrieval-Augmented Generation (RAG).

Professional Growth

- Learned to adapt quickly to emerging technologies.
- Developed a mindset of continuous learning and self-improvement.
- Understood current industry trends in Full Stack Development and Artificial Intelligence.
- Strengthened analytical thinking and technical problem-solving skills.

4.2 Challenges Faced

- Transitioning from basic programming to JavaScript concepts such as asynchronous programming and event-driven execution.
- Understanding the rapidly evolving field of Agentic AI and its ecosystem.
- Learning advanced AI frameworks such as LangChain and LangGraph.
- Balancing multiple learning objectives within the internship duration.
- Connecting theoretical concepts with real-world applications.

4.3 Results

Despite the challenges encountered, the internship was completed successfully, resulting in the following measurable outcomes:

- Successfully completed two industry-recognized professional certifications: JavaScript Essentials 1 (Cisco Networking Academy) and Agentic AI Certified Foundations Associate (Oracle University).
- Strengthened core programming skills in JavaScript.
- Expanded knowledge of modern Full Stack technologies, including React.js and Node.js.
- Gained a foundational understanding of Agentic AI, AI agents, and intelligent automation concepts.
- Improved problem-solving, analytical thinking, and continuous learning skills.

CHAPTER 5 CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The Summer Internship provided valuable, industry-aligned exposure to both modern web development and emerging Artificial Intelligence technologies. Through the Cisco Networking Academy's JavaScript Essentials 1 course and Oracle University's Agentic AI Certified Foundations program, the intern was able to strengthen core programming skills, gain practical knowledge of the MERN Stack, and develop a working understanding of Agentic AI, LLMs, and AI agent frameworks such as LangChain, LangGraph, and RAG. The successful completion of two industry-recognized certifications validates the skills and knowledge acquired during this period and marks a significant step toward becoming industry-ready in the domains of Full Stack Development and Artificial Intelligence.

5.2 Future Scope

- Building complete, deployable full-stack projects using the MERN Stack.
- Exploring advanced Agentic AI concepts, including multi-agent orchestration and tool-augmented reasoning.
- Gaining deeper, hands-on experience with LangChain, LangGraph, and Retrieval-Augmented Generation (RAG) pipelines through independent projects.
- Pursuing further certifications in cloud computing and AI to complement the skills gained during this internship.
- Applying the combined knowledge of Full Stack Development and Agentic AI toward internship or industry roles involving intelligent, AI-powered web applications.

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- [7] LangChain Documentation. Available: <https://www.langchain.com>
- [8] LangGraph Documentation. Available: <https://www.langchain.com/langgraph>

APPENDIX 1 CERTIFICATES

A1.1 JavaScript Essentials 1 Certificate

Awarded to: Mr. Paarth Gupta

Issued for: Successful completion of JavaScript Essentials 1, offered by Networking Academy through the Cisco Networking Academy program.

Issued by: Lynn Bloomer, Director, Cisco Networking Academy

Completion Date: 24 Jul 2026

Certificate ID: f100b158-f36d-4ef7-b937-bba025f70d3c

A1.2 Oracle Certified Foundations Associate Certificate

Awarded to: Paarth Gupta

Credential: Agentic AI Certified Foundations Associate

Issued by: Gary N. Miller, Customer Success Officer, EVP CSS, Oracle University

Date: August 04, 2026

Certificate No.: 103504149AAI26OFA